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B. E. (Fourth Semester) Examination, April-May 2016

(New Scheme)

(IT Branch)

DATA STRUCTURES and ALGORITHM ANALYSIS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all questions. Part (a) of each questions is compulsory & carry 2 marks. Solve any two part out of (b), (c) & (d) from each questions & carry 7 marks.*

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Unit-I

1. (a) What do you mean by asymptotic notation? 2

(b) Explain different types of algorithm in detail. 7

(c) Explain master method for solving recurrence relation. Consider the following recurrence

$$T(n) = 4T(n/2) + n$$

Solve this using master method. 7

(d) Explain representation of two dimensional array in memory. Consider the 25 × 4 matrix array A. Suppose Base $(A) = 200$ and there are $w = 4$ words per memory cell. Find effective address for $A[12, 3]$ using row-major order. 7

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Unit-II

2. (a) What is Stack? 2

(b) Write the algorithm to transform infix expression into postfix expression. Convert following expression into postfix notation. 7

$$A * B + C/D$$

(c) Explain circular queue, Do-Queue, and priority queue with example. 7

(d) Write an algorithm to insert an element in the linked list : 7

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(i) At the beginning

(ii) At the known location

(iii) At the end

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Unit-III

3. (a) Define divide and conquer strategy. 2

(b) Suppose array A has 12 elements

44, 33, 11, 55, 77, 90, 40, 60, 99, 22, 88, 66

sort above array using quick sort algorithm. 7

(c) Write binary search algorithm and explain with example. 7

(d) What is hashing? Explain collision resolution technique in hasing. 7

Unit-IV

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4. (a) What is m-way search tree? 2

(b) Write tree traversal algorithm (recursive). A binary tree T has 9 nodes. The inorder and preorder traversals of T yield the following sequences of nodes.

Inorder : E A C K F H D B G

Preorder : F A E K C D H G B

Draw the tree T. 7

(c) Wrie algorithm to search an element in binary search tree.

Create binary search tree for the following given numbers

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40, 60, 50, 33, 55, 11 7

(d) Explain LL and RR rotation of AVL tree with example. 7

Unit-V

5. (a) Define Graph. 2

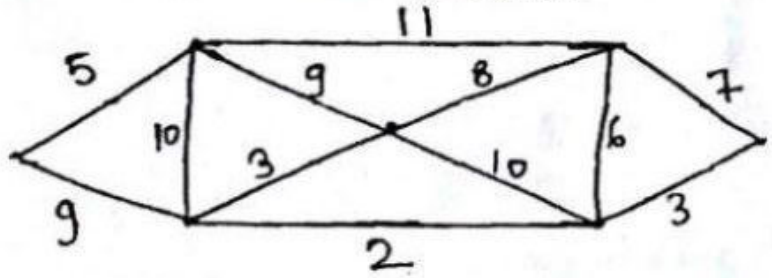
(b) Explain the Breadth first search algorithm with example. 7

(c) Explain sequential representation of graph using : 7

(i) Adjacency matrix

(ii) Adjacency list

(d) What is Minimum Spanning Tree? Create minimum spanning tree for the following graph. 7



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